Walmart Customer Segmentation

Using K-Means Clustering

Dataset: Walmart_customer_purchases.csv
Algorithm: K-Means Clustering (Unsupervised Learning)
Goal: Group customers by behaviour and suggest targeted actions
Report Date: 05/09/2026

Business Problem

Walmart has millions of clients visiting its stores. However, these clients cannot be considered the same since some spend a lot of money, while others only spend occasionally. Also, there are people of different ages visiting Walmart; some are young people, while others are old people. Sending the same offer to all the clients would lead to losses for Walmart because it will be using unnecessary funds.
This paper seeks to develop an algorithm for segmenting Walmart's customers into various groups according to their behaviors and sending them appropriate offers.

Data Overview

The dataset being used here is Walmart_customer_purchases.csv. Every row is a transaction performed by one of the customers. The important columns are
Customer_ID, Purchase_Amount, Age, Gender, Category, Rating, Repeat_Customer

Step-by-Step Process

Step 1: Load and Explore the Data

Data loading was done using pandas. First and last five rows were reviewed, dimensions of data were verified, columns present were identified, and presence of missing data was checked. There were no missing values found, hence no cleaning required.

Step 2: Build Customer Metrics

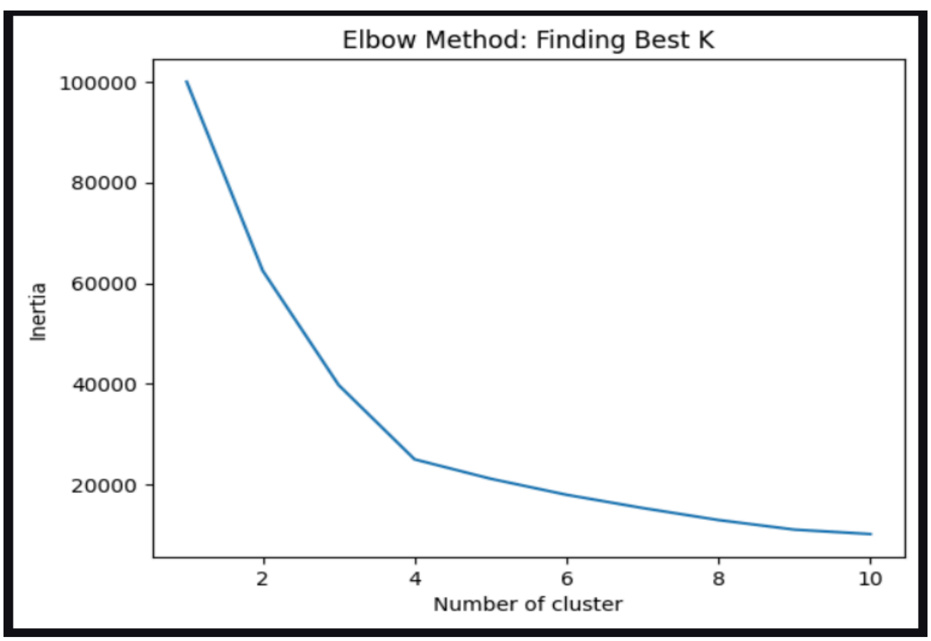
The raw dataset had entries per purchase. This was categorized according to the Customer_ID, with three critical metrics determined per Customer_ID:
Total Spend - Total money spent by the client on all the purchases.

Frequency - The number of times the purchase was made by the client.
Average - Average amount of money spent per transaction.
Other factors such as age, gender, category, rating, and repeat customers were also retained.

Step 3 - Encode Text into Numbers
K-Means only works with numbers, not text. So converted text columns: Gender: Male = 1, Female = 0 Category: each product type was given a unique number automatically. Repeat Customer: Yes = 1, No = 0

Step 4 - Scale the Data
Different columns have very different ranges. For example, Total Spend can be in the thousands while Age is only around 20 to 70. If it's not properly scaled, K-Means will think Total Spend is more important just because its numbers are bigger

Step 5 - Find the Best K using the Elbow Method
Rather than guessing the optimal number of clusters to use, ran the K-Means algorithm for K values between 1 and 10 and plotted the inertia. The plot revealed sharp drops at K = 2 and K = 3, and moderate drops at K = 4. The improvement beyond K = 4 was marginal. Therefore, K = 4 is the selected value.



Step 6 - Train the Final Model

Trained KMeans with n_clusters=4 and random_state=42 on the scaled data. The model assigned every customer to one of 4 clusters. Then added the cluster label back to the customer dataframe

Cluster Results

After training, the average numbers in each cluster to understand who is in it.

Deep Cluster Analysis

This output shows the most bought category, most used payment method, and loyalty rate for each cluster. This deeper analysis is what made the recommendations data backed instead of just guesses.

```
Cluster      : Budget Youth
Top Category : Clothing
Top Payment  : Credit Card
Loyalty Rate : 50.6%
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Cluster      : Budget Seniors
Top Category : Beauty
Top Payment  : Debit Card
Loyalty Rate : 50.1%
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Cluster      : High Value Youth
Top Category : Electronics
Top Payment  : Debit Card
Loyalty Rate : 50.6%
----------------------------------------

Cluster      : High Value Seniors
Top Category : Electronics
Top Payment  : Cash on Delivery
Loyalty Rate : 50.7%
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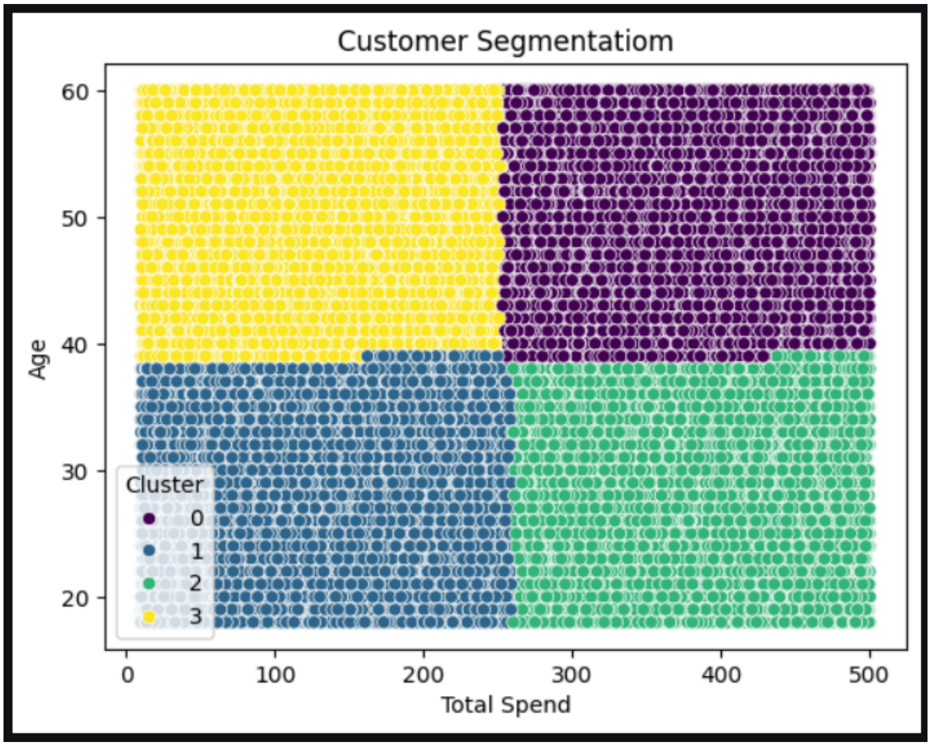
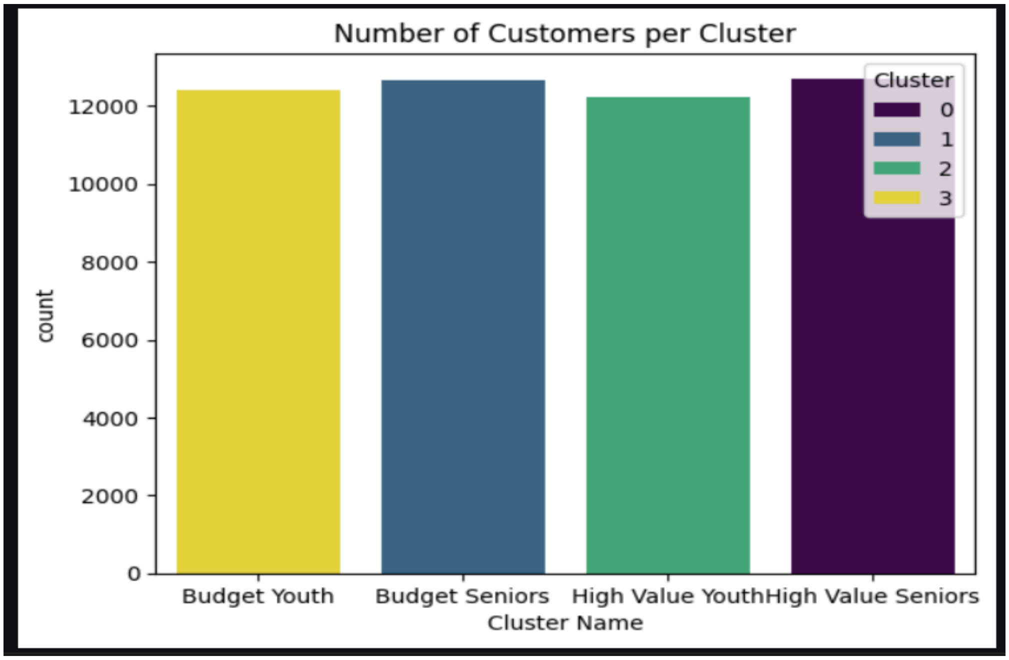
Scatter Plot

Each dot is one customer. The colour shows which cluster they belong to. There are 4 separate groups forming based on Age and Total Spend.



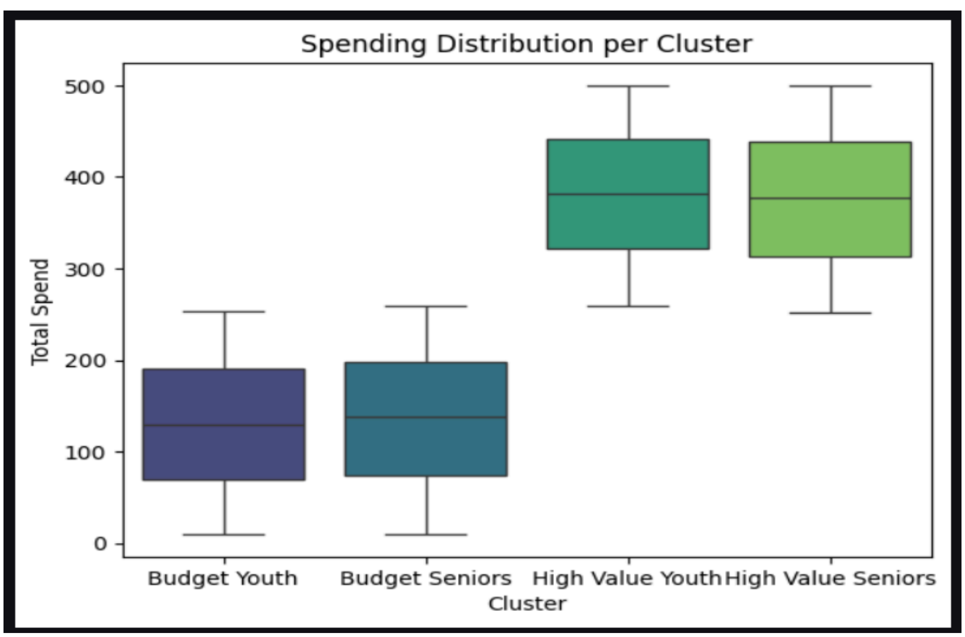
Count Plot

This chart shows how many customers are in each cluster. All four groups are nearly equal in size around 12,000 to 13,000 customers each. This is a very healthy result because it means the algorithm found balanced groups and no single cluster is dominating the data.



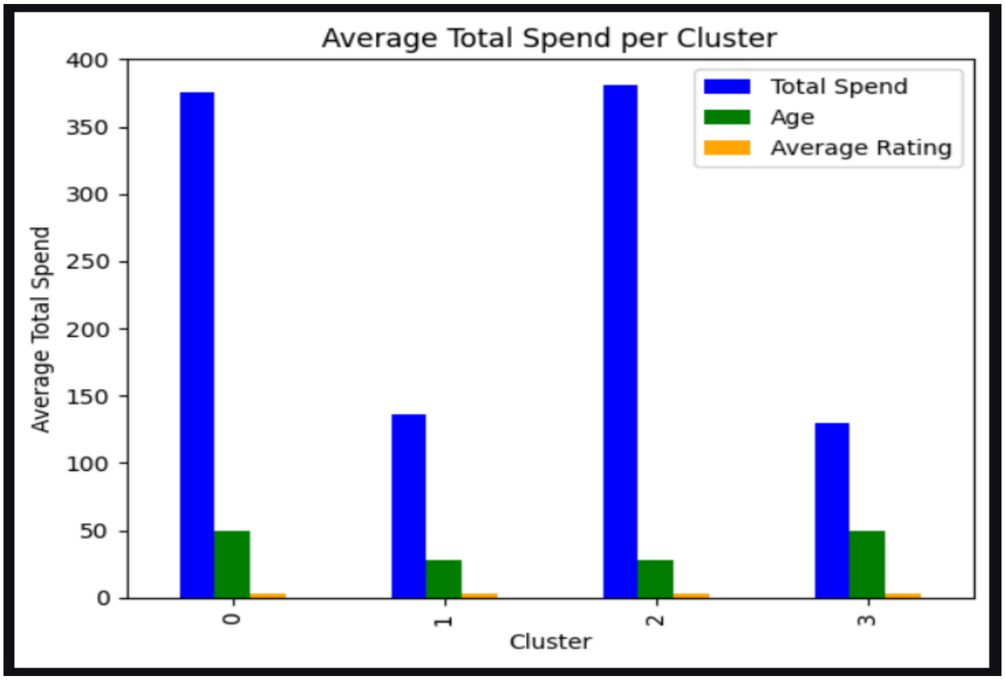
Box Plot

The box plot confirms the cluster names are correct. Budget Youth and Budget Seniors both sit between $0 and $250, proving they are low spenders. High Value Youth and High Value Seniors both sit between $250 and $500, proving they are high spenders. The middle line inside each box shows the average spend per cluster.



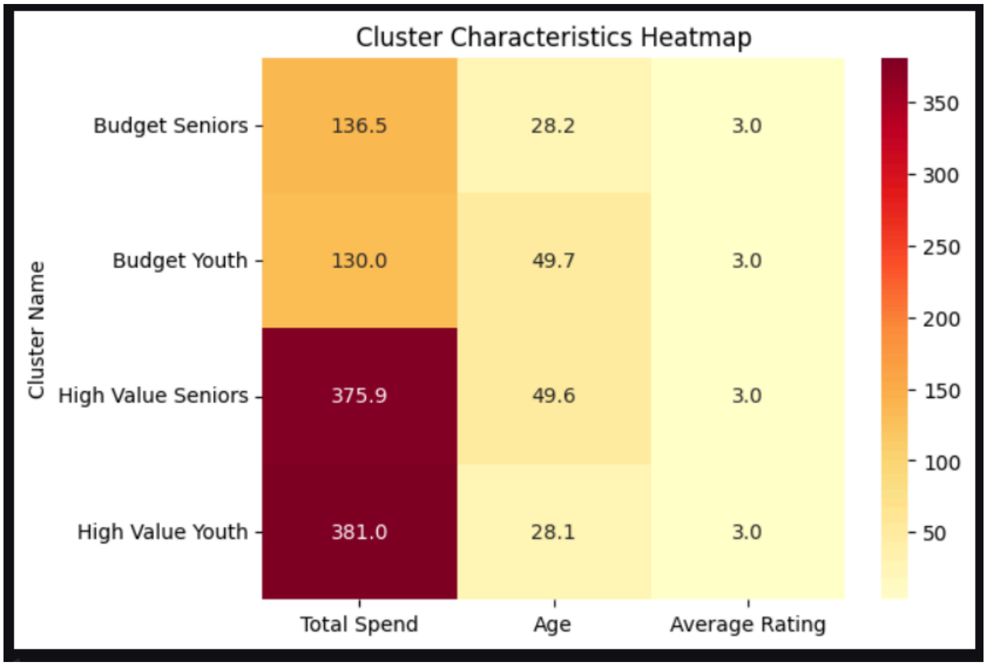
Bar Plot

Each box shows the spending range within a cluster. A tall box means customers in that group have varied spending. A short box means they all spend similarly.



Heatmap

The heatmap gives a quick visual summary of all clusters at once. Darker colours mean higher values. It helps to instantly spot which clusters have high spending and which have low spending.



Recommendations

According to the results of cluster analysis, a suitable recommendation was made for each category of customers. For High Value Seniors, loyalty points were awarded, while Electronics with high value were recommended. In the case of Budget Seniors, coupons for discounts on Beauty items were provided every week. The trending electronics with debit card cash back was recommended to High Value Youth.

Conclusion

Customers at Walmart include many people with various demands and budgeting patterns. Four customer clusters were found in the study, and a suitable recommendation was made for each customer cluster in accordance with their actual behavior and payment methods. This method will enable Walmart to boost customer satisfaction, encourage repeat purchases, and cut down marketing expenses spent on wrong offers.